TENSORFORGE 2.0

Phase 2 — MVP Stage

Student Rules & Guidelines Booklet

Key Dates

Phase 2 starts today at 6:00 PM. Submission deadline: 10th October 2026, 06:00 PM. No extensions will be granted — plan your build and testing time accordingly.

1. Overview

In this stage, one shared problem statement is released to all shortlisted teams: **a customer support ticket classification and routing system**. You will build a working prototype, host a batch inference endpoint for evaluation, and submit a repository, a video walkthrough and a short report.

2. Attachments Provided

1. tensorforge-phase2-openapi-v2.yaml — the OpenAPI 3.0 contract that your service must follow.
2. schemas/ — JSON Schemas (draft 2020-12) for every request and response, plus a single bundle file (tensorforge-schemas.json). Use these to validate your service locally. They also encode consistency rules such as the fixed category-to-team mapping.

3. API Specification — What It Covers

Endpoint	Purpose
POST /predict	Single-ticket prediction
POST /predict/batch	Batch prediction for 1-100 tickets (synchronous)
POST /batch/jobs	Submit a large evaluation job (more than 2,000 tickets, up to 5,000 per job)
GET /batch/jobs/{job_id}	Poll job status (asynchronous)
GET /batch/jobs/{job_id}/results	Fetch results once the job is complete
GET /health	Public health check endpoint

All prediction and job endpoints require API key authorization. GET /health is public.

Latency is not part of the evaluation. Scoring is based on correctness, schema compliance and robustness.

4. API Key — Handling Rules

Protect your API key

Each team leader will be given an unique API key by email. Do not share it or lose it. Judges will use this exact key during evaluation, so it must remain valid and working for the full evaluation window. Engineer your authentication so it works reliably with this key under load.

5. Final Deliverables

1. Complete end-to-end solution: a demo site, dashboard or similar front end. Full freedom in design and presentation — share a link to the live solution.
2. Batch inference endpoint: hosted and authenticated, used to run the judges' holdout dataset. Because evaluation is automated, it must strictly follow the API specification — field names, enum values, status codes and error formats.
3. GitHub repository: the full project code, runnable and testable locally. The container must start with a single "docker run -p 8000:8000 -e API_KEY=<key> <image>" command and must not need internet access at runtime.
4. Video (15 minutes): a walkthrough of the solution, the architecture, and how it was evaluated.
5. Report (max 5 pages): design decisions, evaluation scores and findings.

6. Approach Guidance — Read Carefully

Do not wrap an LLM for classification

Solutions that simply call an LLM to perform the ticket classification will be discouraged and scrutinized. Judges will look closely at each team's model training approach. Your video, repository and report must clearly show how your model was trained, validated and improved.

You have full freedom to choose your solution approach beyond this restriction — classical ML, fine-tuned models, ensemble methods and similar approaches are all welcome, as long as your own training and validation process is clearly demonstrated.

7. Pre-Submission Checklist

- Service validated locally against the provided JSON Schemas
- All prediction and job endpoints require the API key; /health remains public
- Docker container starts with the single documented run command and needs no internet access at runtime
- Batch inference endpoint hosted, authenticated, and tested end-to-end
- Live demo link working and accessible
- Video (15 minutes, YouTube unlisted) covers the solution, architecture and evaluation approach
- Report (max 5 pages) covers design decisions, evaluation scores and findings
- Model training, validation and improvement process clearly documented — not just an LLM call
- API key kept private and working for the full evaluation window

8. Support

Mentor check-ins are available throughout Phase 2. Reach out to the organizing committee through the official TensorForge communication channel for any questions on the specification, schemas or submission process.

Download the Resources & Dataset : - Click Here

TensorForge 2.0 Organizing Committee — IEEE Computer Society KDU